Sujal Rai

+91 7898152310 | raisujal205@gmail.com | linkedin.com/in/sujalrai | Portfolio
Bhopal, Madhya Pradesh, India

Summary

I am a Computer Science and Engineering undergraduate specializing in Artificial Intelligence and Machine Learning, with hands-on experience building full-stack applications using the MERN stack. I enjoy turning ideas into practical software, from web platforms to AI-powered tools. I bring a curious, analytical approach to problem-solving and am eager to keep learning while contributing to useful technology projects.

Skills

Languages: Python, JavaScript, Core Java, SQL

Frontend: HTML, CSS, Tailwind, EJS, React

Backend: Node.js, Express.js

Databases: MongoDB, SQL

Core Subjects: DBMS, Object-Oriented Programming (OOP)

Tools & Platforms: Git, GitHub, Render, VS Code, Vercel

Projects

TravelNest
(EJS, JavaScript, Node.js, Express.js, MongoDB) Link

- Built a full-stack Airbnb-style application featuring user authentication, property listings, image uploads, and booking functionality using MongoDB and Express.js.
- Designed and implemented **user authentication**, property listing, and booking modules with secure session handling.
- Utilized EJS templating for a responsive frontend and Node.js/Express.js for scalable RESTful backend services.

ExamNotes AI
(React, JavaScript, Node.js, Express.js, MongoDB) Link

- Built a full-stack AI-powered study tool that generates structured, exam-focused notes tailored to a student's class level and exam type. It also supports revision summaries, quizzes, diagrams, charts, saved note history, and PDF downloads.
- Integrated Groq to generate structured notes with optional Mermaid diagrams and Recharts visualizations; added Markdown rendering and PDF export.
- Implemented Google sign-in, JWT cookie-based authentication, MongoDB note history, and Stripe checkout for credit purchases.

Pothole Detection & Reporting Platform
(React.js, JavaScript, Node.js, Express.js) Link

- Developed a full-stack pothole reporting platform with a React.js frontend and a Node.js/Express.js REST API, enabling citizens to report road damage.
- Created an **admin dashboard** to track total, open, in-progress, and fixed reports at a glance.
- Designed a recent reports view that lets administrators review reported road issues and their current statuses.
- Integrated a **map view** to visualize pothole report locations and help identify problem areas.

Education

Oriental Institute of Science and Technology Aug. 2023 – Present
B.Tech. – Computer Science and Engineering (Artificial Intelligence & Machine Learning) Bhopal, M.P.

- **CGPA:** 7.0
- **Relevant Coursework:** Data Structures, Algorithms, DBMS, OOPs

Shantiniketan School
12th – Science Stream Bhopal, M.P.

- **Percentage:** 80%

Certifications

- **Introduction to Data Science** – Infosys
- **Introduction to Cybersecurity** – Cisco
- **Mastering Python** – Udemy